

CMAD: Correlation-Aware and Modalities-Aware Distillation for Multimodal Sentiment Analysis with Missing Modalities

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Abstract

*Multimodal Sentiment Analysis (MSA) enhances emotion recognition by integrating information from multiple modalities. However, multimodal learning with missing modalities suffers from representation inconsistency and optimization instability, leading to suboptimal performance. In this paper, we introduce **Correlation-Aware and Modalities-Aware Distillation (CMAD)**, a unified framework designed for MSA under varying missing-modality conditions. Specifically, CMAD comprises two key components: (1) **Correlation-Aware Feature Distillation (CAFD)**, which enforces multi-level representation alignment by preserving both feature similarities and high-order correlation structures between teacher and student models, and (2) **Modality-Aware Regularization (MAR)** employs an adaptive weighting strategy guided by modality difficulty, enabling a curriculum learning paradigm to stabilize the training process. Extensive evaluations on five datasets show that CMAD consistently outperforms existing methods, achieving average performance improvements of 1.0% on MOSEI, 4.4% on IEMOCAP, 1.9% on MUSARD, 0.5% on UR-FUNNY and 1.9% on CHERMA. Code is available at: <https://github.com/YetZzzzzz/CMAD>.*

1. Introduction

Multimodal Sentiment Analysis (MSA) aims to predict the sentiment of utterances by integrating information from multiple modalities, typically language, audio, and video [22, 45]. However, real-world applications often encounter missing modalities due to data collection limitations [25] and privacy concerns [13]. Traditional models [43, 51] assume full availability of modality, resulting in significant performance drops when modalities are missing.

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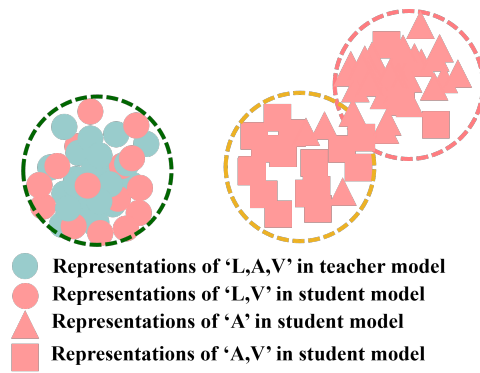


Figure 1. Illustration of the challenges in KD: The representation inconsistency of different modalities (e.g. ‘L,A,V’ and ‘A’) is large, with language (‘L’) offering the most discriminative signal.

Existing approaches to handling missing modalities can be broadly categorized into two main types based on their training paradigms: *reconstruction-based* and *knowledge distillation (KD)-based*. For **reconstruction-based** methods [20, 29, 37, 39, 40, 49], current approaches focus on reconstructing missing modalities from available ones using generative models, such as diffusion-based [31, 40] or flow-based techniques [39]. While effective, these methods often produce modality-specific rather than globally consistent representations. Additionally, in a MSA setting with m modalities, these approaches often require separate training per missing modality combination, resulting in training up to $2^m - 1$ different models to cover all possible combinations, making them impractical for real-world applications where managing such a large number of models is difficult. In contrast, **knowledge distillation (KD)** offers a promising alternative by transferring knowledge from a teacher model trained on all modalities to a student model operating under limited modalities [7, 15–17, 32, 38, 42, 44].

While KD provides a more scalable solution, it introduces two major challenges: (1) **Representation Inconsis-**

tendency: Missing modalities lead to discrepancies between the feature spaces of the teacher and student models, which can be seen in Fig. 1. A naïve approach, such as minimizing the mean squared error (MSE) between their representations, fails to capture high-order semantic relationships, making conventional knowledge transfer ineffective. Furthermore, without explicitly aligning the relational structure within the teacher’s embedding space, semantic similarities among instances may be lost, leading to distorted feature representations. **(2) Optimization Instability:** Modalities contribute to sentiment analysis with varying levels of reliability, with language typically offering the most discriminative signal in Fig. 1. Consequently, easier modality combinations (e.g., language and video) tend to dominate the learning process, converging faster than more challenging configurations (e.g., audio-only). This imbalance can result in suboptimal local minima, where the model underutilizes difficult modalities, leading to poor generalization.

To address these challenges, we propose Correlation-Aware and Modalities-Aware Distillation (CMAD), a novel KD framework designed to enhance knowledge transfer under missing-modality scenarios. CMAD consists of two key modules:

- **Correlation-Aware Feature Distillation (CAFD):** This module ensures multi-level alignment by complementing traditional feature-matching techniques with correlation alignment loss, which preserves high-order semantic relationships between instances.
- **Modality-Aware Regularization (MAR):** Inspired by curriculum learning [2], MAR dynamically adjusts learning weights, gradually shifting focus from easier to more challenging modality combinations. This progressive strategy prevents dominant modalities from overshadowing others, stabilizes training, and ensures balanced optimization across all configurations.

By jointly optimizing CAFD and MAR, CMAD effectively addresses representation inconsistency and training imbalance, ensuring robust knowledge transfer and stable optimization. Our key contributions are as follows:

- We present CMAD, a unified framework that generates unified representations across various missing modalities.
- We develop a sample-wise consistency mechanism (CAFD) that improves representation robustness by ensuring feature matching between student-teacher pairs and correlation alignment across samples between student-teacher and teacher-teacher representations.
- We propose an adaptive weighting strategy (MAR) to handle prediction difficulties across different modality combinations, enhancing representational balance between the student and teacher models.
- We conduct comprehensive experiments on five datasets, demonstrating that CMAD significantly outperforms existing methods.

2. Related Work

2.1. MSA with Missing Modalities

MSA uses information from language, video, and audio modalities of an utterance to determine its sentiment [21, 23, 34, 46]. However, in practical applications, some modalities may be missing due to factors like limitations in data collection methods [25], which severely impact the performance of the traditional models [10, 18, 23, 24, 50].

Current solutions can generally be divided into two categories: reconstruction-based and KD-based methods. Reconstruction-based methods tend to reconstruct the absent modalities using information from the available ones [8, 20, 26, 29, 41, 49], using sophisticated decoders [20, 41], diffusion models [31, 40], or flow-based generative models [6, 39]. For example, MPLMM [8] utilizes three prompt types—generative, missing-signal, and missing-type—to reconstruct missing modalities. These prompts are used to reconstruct the missing modalities with available modalities. The KD-based models are introduced in Section 2.2.

Despite improving robustness, these methods face notable limitations. They often require separate, customized models for each modality combination, resulting in up to $2^m - 1$ different reconstruction strategies for m modalities. Managing so many models in real-world applications is not practical. In contrast, CMAD ensures multi-level alignment by complementing traditional feature-matching techniques with correlation alignment loss, which preserves high-order semantic relationships between instances, and adapts to different modality combinations with varying representational strengths, thus handling all missing-modality scenarios within a single model.

2.2. Knowledge Distillation

Knowledge distillation (KD) is designed to transfer knowledge from a strong teacher model to a weaker student model [11, 16, 17, 19, 35, 42, 48]. Recently, researchers have adapted KD to incomplete multimodal learning [16, 17, 42]. In these approaches, the teacher model is trained with complete modality data, while the student model learns from data with missing modalities. For example, MMANet [42] transfers correlations from the teacher model to the student model, aligning student-student correlations with teacher-teacher ones.

While these methods demonstrate effectiveness, they face significant limitations. First, the variable modality availability can lead to considerable representation inconsistency between the student and teacher representations. Second, these methods often ignore the varying difficulty of different modality combinations and samples, resulting in suboptimal local minima. In contrast, our proposed CMAD addresses these issues by ensuring multi-level alignment, while adapting to diverse modality combinations.

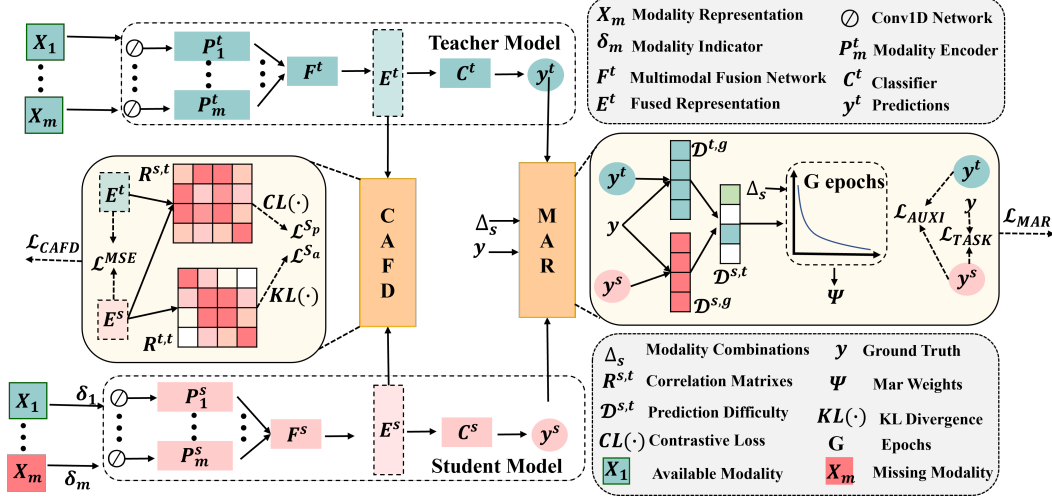


Figure 2. Overview of the proposed CMAD framework. It consists of a student model, a teacher model, and two key modules: Correlation-Aware Feature Distillation (CAFD) and Modalities-Aware Regularization (MAR). CAFD ensures feature matching between student-teacher pairs and correlation alignment across samples between student-teacher and teacher-teacher representations, while MAR dynamically adjusts the weight of each modality combinations based on its difficulty.

3. Method

3.1. Problem Definition

Given the multimodal dataset $\mathcal{D}$ with $m = 3$ modalities (i.e. language, audio and video), the input for each batch of size B is represented as $x = \{X_1, X_2, \dots, X_m\}$. For each modality i , $i \in \{1, 2, \dots, m\}$, $X_i \in \mathcal{R}^{B \times T_i \times D_i}$ represents the modality features, with T_i indicating the its temporal length and D_i representing its dimensionality. In practical scenarios, some modalities may be missing, and in such cases, the missing modality is replaced by a vector of zeros. To track the availability of each modality, we introduce a binary indicator vector $\Delta_b = \{\delta_1, \dots, \delta_m\}$, where $\delta_i = 1$ indicates that the i -th modality is available, and $\delta_i = 0$ indicates its absence [42]. For an input with missing modalities, the corresponding representation becomes $x_{in} = \{\delta_1 X_1, \delta_2 X_2, \dots, \delta_m X_m\}$. The goal is to learn robust and consistent representations that can accurately predict sentiment using incomplete input data x_{in} .

3.2. CMAD Overview

As shown in Fig. 2, CMAD includes two modules: the Correlation-Aware Feature Distillation (CAFD) module (introduced in Section 3.3) and the Modalities-Aware Regularization (MAR) module (introduced in Section 3.4). Besides, it comprises a teacher model pre-trained using full modalities and a student model trained with limited modalities, as proposed in prior works [17, 42]. For a batch input of size $x = \{X_1, X_2, \dots, X_m\}$, some modalities (e.g. X_j, X_k) of samples in student model are randomly discarded by setting $\delta_j = 0$ and $\delta_k = 0$ in Δ_b to simulate real-world scenarios.

The processed inputs are passed through a Conv1D layer to capture temporal information, making all modalities X_i uniform in shape, with $X_i \in \mathcal{R}^{B \times T \times D}$, where T and D are hyper-parameters representing common modality length and dimension. Each processed modality is then fed into a modality encoder P_j^w , where $j \in \{1, 2, \dots, m\}$ denotes the specific encoder for modality j . We adopt the Perceiver architecture [12, 21] as our modality encoder. The encoded modality representations are then fused through a fusion network F^w , producing the fused embeddings E^w . Here, we use a transformer encoder [36] followed by a linear layer [10] as the F^w for fusion. These fused embeddings from both the student (E^s) and teacher models (E^t) are then input into the CAFD module, where the student learns sample-wise feature and correlation consistency from the teacher model. Finally, the classifier C^w predicts the final logits y^w . The logits of both models, the ground-truth labels y , and the modality combinations of samples in current batch are then passed to the MAR module. This module dynamically adjusts weights according to different modality combinations. Here $E^w \in \mathcal{R}^{B \times D}$, $y^w \in \mathcal{R}^{B \times k}$, where $w \in \{t, s\}$ denotes the student model (s) or teacher model (t). k represents the number of classes and the ground truth $y \in \mathcal{R}^{B \times k}$. To simplify the representation of modality combinations, we map the binary indicator vector Δ_b to a scalar value. For example, when three modalities are present, the binary vector $[1, 1, 1]$ would be converted to the scalar ‘7’ by computing ‘ $1 \times 2^0 + 1 \times 2^1 + 1 \times 2^2$ ’. This scalarized modality combination is denoted as Δ_s . Thus the modality combination in the batch is $\Delta_s \in \mathcal{R}^B$. Further details on P_j^w and F^w can be found in Supplementary Material.

3.3. CAFD

Current methods struggle to maintain robustness due to the key challenge that the student model typically has access to only a subset of modalities, while the teacher model uses all modalities. The student model’s missing modalities can disrupt the learning process, especially when dominant modality is absent, causing misalignment in feature and correlation consistency [17, 20]. To address this gap, we introduce the Correlation-Aware Feature Distillation (CAFD) mechanism, which ensures feature matching between student-teacher pairs and correlation alignment across samples between student-teacher and teacher-teacher representations.

Firstly, the sample-wise feature consistency between student (E^s) and teacher representations (E^t) is calculated using the MSE loss function:

$$\mathcal{L}_i^{MSE} = \|E_i^t - E_i^s\|_2, \quad (1)$$

where $\|\cdot\|_2$ denotes the MSE loss, and i denotes the i -th sample. To ensure student-teacher and teacher-teacher correlations alignment, the correlations between teacher-teacher representations (denoted as $R^{t,t} \in \mathcal{R}^{B \times B}$) and student-teacher representations (denoted as $R^{s,t} \in \mathcal{R}^{B \times B}$) are calculated using cosine similarity [17, 42]:

$$R_{i,j}^{w,t} = \frac{E_i^w \top E_j^t}{\|E_i^w\|_2 \|E_j^t\|_2}. \quad (2)$$

Here i and j denote the indices in E^w and E^t , respectively. Since the representations from both models correspond to the same samples, the correlations for identical samples should be maximized. To do this, we introduce positive pair correlation loss $\mathcal{L}^{S_p}$ based on contrastive learning [30]. The absolute distance between the two correlation matrices, $R^{t,t}$ and $R^{s,t}$, is computed and the overall correlation for identical samples is then normalized by the total correlations of all samples as follows:

$$\mathcal{L}^{S_p} = CL(E^s, E^t) = \frac{\text{sum}(\text{diag}(|R^{t,t} - R^{s,t}|))}{\text{sum}(|R^{t,t} - R^{s,t}|)}, \quad (3)$$

where $|\cdot|$ denotes the element-wise absolute error, $\text{diag}(\cdot)$ denotes the diagonal elements of the matrices and $\text{sum}(\cdot)$ denotes the sum of all matrix elements. The $CL(\cdot)$ denotes the contrastive learning loss.

While for the sample-wise correlation consistency across all samples, we hope the similarities between each student representation and every teacher representation closely resemble that between the corresponding teacher representation and every teacher representation, which can be formulated as distribution discrepancy problem. To address this, we introduce the all pair samples correlation loss $\mathcal{L}^{S_a}$ by employing Kullback-Leibler (KL) divergence, as referenced in previous studies [16, 42], and incorporating a temperature parameter τ to smooth the learning process:

$$\begin{aligned} \mathcal{L}_i^{S_a} = & \frac{1}{2} \sum_{j=1}^B [KL(\log(\sigma(\frac{R_{i,j}^{t,t}}{\tau})), \sigma(\frac{R_{i,j}^{s,t}}{\tau})) \\ & + KL(\log(\sigma(\frac{R_{i,j}^{s,t}}{\tau})), \sigma(\frac{R_{i,j}^{t,t}}{\tau}))], \end{aligned} \quad (4)$$

where $KL(\cdot)$ is the KL divergence, $\sigma(\cdot)$ denotes the softmax function for normalization, τ is the hyper-parameter, and $i \in \{1, \dots, B\}$ denotes the i -th representation.

Finally, the overall loss of CAFD is computed through:

$$\mathcal{L}_{CAFD} = \mathcal{L}^{S_p} + \frac{1}{B} \sum_{i=1}^B \mathcal{L}_i^{S_a} * \mathcal{L}_i^{MSE}. \quad (5)$$

3.4. MAR

Different modalities contribute with varying levels of reliability to sentiment analysis. Easier modality combinations, such as language and video, tend to dominate the learning process and converge faster than more challenging ones, like audio-only. This imbalance can lead the model to rely heavily on easy combinations while underutilizing difficult modalities. To address this, we propose the Modalities-Aware Regularization (MAR) module, which dynamically adjusts the weight of each modality combination based on its difficulty. Recognizing that the difficulty of a modality combination is influenced not only by the modalities involved but also by the difficulty of individual samples. Relying solely on the student model’s predictions to determine a combination’s difficulty is problematic, as some samples are inherently more challenging to classify than others. MAR mitigates this issue by eliminating the influence of sample difficulty.

Since the teacher model has access to all modalities, while the student model operates under missing modality conditions. The student model should first acquire sufficient knowledge from each modality combination before weighting them based on difficulty. Inspired by curriculum learning [2], MAR follows a two-stage process. (1) Initial Learning Phase (Epochs $E \leq G$): All modality combinations are treated equally in training. During this phase, easier modality combinations naturally dominate the learning process. Simultaneously, we record sample difficulty and the performance of each modality combination. (2) Dynamic Re-weighting Phase (Epochs $E > G$): At the end of epoch G , we use the recorded values to compute the MAR weights Ψ for all modality combinations. From epoch $E > G$ onward, MAR dynamically re-weights the loss weights for modality combinations with computed Ψ .

To eliminate the sample difficulty, we introduce a difficulty metric, which is quantified using the loss between the logits y^w and the ground truth y . Specifically, for epochs $E \leq G$, we apply a task-specific loss function $\mathcal{L}_{TASK}$ (defined in Eq. 11) to compute the sample difficulty as:

$$\mathcal{D}_i^{w,g} = \mathcal{L}_{TASK}(y_i^w, y_i), \quad (6)$$

where i denotes the i -th sample and $w \in \{t, s\}$ denotes the student model (s) or teacher model (t). A higher value of $\mathcal{D}_i^{t,g}$ indicates a harder sample, as it is more difficult to predict even with all modalities. The difficulty of each modality combination, after eliminating the sample difficulty, is computed by comparing the teacher's difficulty $\mathcal{D}_i^{s,g}$ with the student's difficulty $\mathcal{D}_i^{t,g}$. In cases where the student model performs better than the teacher (i.e., $\mathcal{D}_i^{s,g} < \mathcal{D}_i^{t,g}$), we set the difficulty difference to zero:

$$\mathcal{D}_i^{s,t} = \begin{cases} 0, & \text{if } \mathcal{D}_i^{s,g} < \mathcal{D}_i^{t,g} \\ \mathcal{D}_i^{s,g} - \mathcal{D}_i^{t,g}, & \text{otherwise} \end{cases}. \quad (7)$$

The difficulties $\mathcal{D}^{s,t}$ and their corresponding modality combinations Δ_s for all samples over the first G epochs are then stored in two dedicated memory banks, $M_l \in \mathcal{R}^{NG}$ and $M_p \in \mathcal{R}^{NG}$, respectively. Here N is the number of all samples in one epoch. These stored values are used to compute the MAR weights $\Psi \in \mathcal{R}^{2^m-1}$ for all modality combinations at the end of epoch G . To do this, we first use a mask matrix to track the modality combination of each sample during the first G epochs. The mask matrix $M_m(i, j) \in \mathcal{R}^{NG \times (2^m-1)}$ is defined as:

$$M_m(i, j) = \begin{cases} 1, & \text{if } M_p(i) = j \\ 0, & \text{otherwise} \end{cases}. \quad (8)$$

Here, $j \in \{1, \dots, 2^m - 1\}$ denotes the j -th modality combination, and $M_p(i)$ corresponds to the modality combination of i -th sample, where $i \in \{1, \dots, NG\}$. A value of $M_m(i, j) = 1$ indicates that the modality combination for i -th sample matches j -th combination. Next, the number of samples $N_m \in \mathcal{R}^{2^m-1}$ for each modality combination is calculated using the mask matrix M_m through $N_m(j) = \sum_i M_m(i, j)$. The weights for all modality combinations, denoted as $W \in \mathcal{R}^{2^m-1}$, are then computed as:

$$W(j) = \begin{cases} 0, & \text{if } N_m(j) = 0 \\ \frac{(M_l \times M_m)(j)}{N_m(j)}, & \text{otherwise} \end{cases}. \quad (9)$$

To ensure these weights are balanced, we normalize them to prevent extreme differences, square them to enhance their distinction, and multiply by a hyper-parameter ρ to generalize to various tasks:

$$\Psi(j) = \rho \left(\frac{W(j)}{\max(W)} \right)^2, \quad (10)$$

where $\max(\cdot)$ means the maximum value in W .

Following common practices in KD [17, 42], we also distill the logits and incorporate the task-specific loss, which are denoted as the auxiliary loss $\mathcal{L}_{AUXI}$ and $\mathcal{L}_{TASK}$.

Specifically, the $\mathcal{L}_{TASK}$ is defined as follows:

$$\mathcal{L}_{TASK}(y^w, y) = \begin{cases} MAELoss(y^w, y), & \text{if } k = 1 \\ CrossEntropy(y^w, y), & \text{if } k > 1 \end{cases}, \quad (11)$$

where k is the number of the classes. While for $\mathcal{L}_{AUXI}$, we follow the approach [17] by utilizing the DKD loss [48] for classification tasks with class numbers $k > 1$, which considers both target and non-target class probabilities:

$$DKD(y^s, y^t) = \alpha KL(b^t || b^s) + (1 - \alpha) KL(\hat{p}^t || \hat{p}^s). \quad (12)$$

Here b^t and b^s denote teacher's and student's binary probabilities of the target class, while $\hat{p}^t$ and $\hat{p}^s$ represent the probabilities among non-target classes for the teacher and student, respectively. For regression tasks with $k = 1$, we employ the MSE loss and introduce a hyper-parameter ϕ to smooth the learning process. Finally, the $\mathcal{L}_{AUXI}$ is defined as follows:

$$\mathcal{L}_{AUXI}(y^s, y^t) = \begin{cases} MSE(\frac{y^s}{\phi}, \frac{y^t}{\phi}) * (\phi)^2, & \text{if } k = 1 \\ DKD(y^s, y^t), & \text{if } k > 1 \end{cases}. \quad (13)$$

The overall loss $\mathcal{L}_{MAR}$ is computed through:

$$\mathcal{L}_{MAR} = \mathcal{L}_{TASK} * M_m^n * \Psi + \gamma \mathcal{L}_{AUXI} * M_m^n * \Psi, \quad (14)$$

where M_m^n is the mask matrix calculated from the current batch, α, γ and ϕ are hyper-parameters. Note that Ψ is computed just once at the end of the epoch G . When $E \leq G$, all values in Ψ are set to 1.0, indicating equal weighting for each modality combination. After epoch G , both $\mathcal{L}_{TASK}$ and $\mathcal{L}_{AUXI}$ are updated using the MAR weights.

Finally, the overall training loss of the CMAD is composed of $\mathcal{L}_{MAR}$ and $\mathcal{L}_{CAFD}$ using a hyper-parameter σ :

$$\mathcal{L}_{total} = \mathcal{L}_{MAR} + \sigma \mathcal{L}_{CAFD}. \quad (15)$$

4. Experiments

4.1. Datasets and Evaluation Metrics

To thoroughly evaluate the generalizability and effectiveness of our proposed CMAD framework, we conduct experiments on a diverse set of multimodal affective computing datasets, covering both multimodal sentiment analysis (MSA) and multimodal emotion recognition (MER) tasks. Specifically, we evaluate CMAD on one MSA dataset (MOSEI [46]), and four MER datasets: IEMOCAP [3], MUS-TARD [4], UR-FUNNY [9], and CHERMA [33]. MOSEI consists of 22,856 video clips, where each clip is labeled with a sentiment score ranging from -3 (strong negative) to 3 (strong positive). To ensure a fair comparison with previous work [8, 40], we converted the regression predictions

Set	Models	Modalities							Avg.
		L	A	V	L,A	L,V	A,V	L,A,V	
Results on MOSEI									
C	MCTN[29]	82.6/82.8	62.7/54.5	62.6/57.1	83.5/83.3	83.2/83.2	63.7/62.7	84.2/84.2	74.6/72.5
	GCNet[20]	83.0/83.2	60.2/60.3	61.9/61.6	84.3/84.4	84.3/84.4	64.1/57.2	85.2/85.1	74.7/73.7
	MPLMM[8]	84.6/84.6	61.8/49.4	62.2/53.2	85.1/85.1	85.2/85.1	63.2/51.0	85.3/85.3	75.3/70.5
	UMDF[16]	85.3/85.2	62.9/59.4	62.1/62.4	85.3/85.3	85.3/85.2	65.4/62.8	85.7/85.6	76.0/75.1
	CorrKD[17]	85.3/85.1	61.9/56.8	63.0/62.5	85.3/85.2	85.6/85.5	64.5/62.3	85.7/85.6	75.9/74.7
U	TEA	86.2/85.9	62.7/49.1	62.7/49.1	86.2/85.9	86.2/85.9	62.7/49.1	86.2/85.9	76.1/70.1
	MMANet[42]	84.6/84.6	62.9/48.5	63.7/63.1	84.8/84.8	85.3/85.2	64.5/60.9	85.3/85.2	75.9/73.2
	CMAD _w	85.8/85.7	63.5/59.6	63.0/62.0	85.7/85.6	86.2/86.1	65.1/64.0	86.0/86.0	76.5/75.6
	CMAD _t	85.4/85.4	63.6/57.8	65.1/62.4	85.6/85.6	86.1/86.1	65.5/63.2	86.0/86.0	76.8/75.2
	CMAD	86.1/ 86.0	63.0/ 60.8	65.7/64.4	86.3/86.2	86.4/86.4	65.6/64.8	86.1/ 86.2	77.0/76.4
Results on IEMOCAP									
C	MCTN[29]	-72.7	-60.2	-57.1	-76.2	-74.4	-62.3	-79.6	-68.9
	GCNet[20]	-76.1	-63.5	-61.6	-79.1	-77.4	-65.4	-82.8	-72.3
	MPLMM[8]	-69.3	-59.7	-57.0	-75.4	-74.5	-67.4	-/-	-67.2
	UMDF[16]	-77.4	-64.9	-62.5	-80.2	-78.6	-66.0	-82.7	-73.2
	CorrKD[17]	-77.7	-65.5	-63.4	-80.0	-77.8	-66.8	-82.8	-73.4
U	TEA	78.0/77.2	77.3/74.7	72.5/69.7	81.2/80.6	77.9/77.2	78.2/76.9	81.5/80.9	78.1/76.7
	MMANet[42]	71.9/63.6	78.3/74.8	73.1/68.9	78.8/76.0	73.8/70.0	79.3/77.1	79.3/77.5	76.4/72.6
	CMAD _w	78.5/77.8	78.9/76.0	73.3/69.9	80.9/80.4	78.8/78.2	78.5/76.7	81.1/80.6	78.6/77.1
	CMAD _t	78.9/78.2	77.6/74.4	73.6/70.6	81.3/80.6	79.2/78.7	77.4/75.5	81.6/81.0	78.5/77.0
	CMAD	79.5/78.7	79.2/76.7	73.6/70.6	81.9/81.3	79.7/79.1	79.0/77.3	81.7/81.2	79.2/77.8
Results on CHERMA									
C	MPLMM[8]	51.5/51.3	65.8/65.3	52.4/52.3	66.3/65.9	64.8/64.8	69.4/69.2	71.2/71.2	63.1/62.9
	UMDF[16]	52.0/51.4	66.9/66.7	55.2/52.8	67.6/67.5	65.4/64.6	70.5/70.4	72.0/71.9	64.2/63.6
	CorrKD[17]	50.4/48.9	66.3/66.0	55.3/53.2	67.2/67.1	65.3/64.3	70.4/70.1	71.8/71.7	63.8/63.0
U	TEA	45.6/40.3	65.1/65.3	53.0/51.9	66.6/66.5	63.4/62.0	69.4/69.5	71.0/71.1	62.0/61.0
	MMANet[42]	51.0/49.1	66.5/66.2	56.0/53.5	67.8/67.6	64.7/63.7	70.3/69.8	72.0/71.8	64.1/63.1
	CMAD _w	52.8/51.9	66.4/66.4	56.8/54.3	67.0/66.9	66.5/65.3	69.7/69.4	71.5/71.2	64.4/63.6
	CMAD _t	53.0/52.1	66.7/66.3	55.7/52.6	68.1/68.0	66.2/65.3	70.5/70.1	72.3/ 72.2	64.6/63.9
	CMAD	53.7/53.5	68.2/68.1	57.0/55.1	69.1/69.1	67.5/67.0	71.1/70.7	72.4/72.2	65.6/65.1

Table 1. Performance comparison under different modality combinations on MOSEI, IEMOCAP and CHERMA datasets. ‘ACC/F1’ is reported. Best performance is bold.

and labels into binary sentiment classes (positive and negative) and reported the accuracy and F1-score. IEMOCAP includes 4,453 samples categorized into four distinct emotion classes. We used the average accuracy and F1-score across the four emotion classes to evaluate the model’s performance. Both UR-FUNNY and MUsTARD are binary classification datasets, consisting of 1,866 and 690 videos, respectively. We evaluated models’ performance using accuracy and F1-score. While CHERMA includes 28,717 Chinese utterances, categorized into seven emotion classes. We used weighted accuracy and F1-score for evaluation.

4.2. Implementation Details

Feature Extraction. For the language features, we used BERT [14] embeddings for the MOSEI, UR-FUNNY,

MUsTARD, and CHERMA datasets, and GloVe [28] embeddings for IEMOCAP. Visual features were extracted using the OpenFace framework [1] for all five datasets, while audio features were extracted using the COVAREP framework [5] for all datasets except CHERMA. For CHERMA, we used the pre-trained wav2vec model [47].

Experiment Setup. All models were implemented using the PyTorch framework [27] and trained on a GTX 3090 GPU. The student model settings were as follows: MOSEI: $G = 20$, $\sigma = 0.1$, $\gamma = 1$, $\phi = 7.0$, a learning rate of $2e-5$ and a batch size of 128. IEMOCAP: $\gamma = 0.7$, $\alpha = 0.9$, $G = 10$, $\sigma = 1$, a learning rate of $1e-3$ and a batch size of 120. CHERMA: $\gamma = 1$, $\alpha = 0.7$, $G = 5$, and $\sigma = 1$, a learning rate of $2e-5$ and a batch size of 400. More details are provided in the Supplementary Material.

4.3. Comparison with State-of-the-art Methods

Comparison. We compare CMAD against several state-of-the-art methods. To ensure a comprehensive evaluation, we assess all possible cases of missing modality within the test set, as illustrated in Table 1. For instance, ‘{L,A,V}’ denotes scenarios where the language, audio and video modalities are all available. Specifically, we evaluate two types of models: customized models, trained separately for each modality combination (denoted as ‘C’) to obtain customized representations, and unified models, which address all missing-modality cases within a single model (denoted as ‘U’) to obtain unified representations. Specifically, for customized models, three reconstruction-based models, including MCTN [29], GCNet [20] and MPLMM [8], and two KD-based models, namely CorrKD [17] and UMDF [16] are compared. Among the unified models, we compare CMAD with the unified MMANet [42] model, which focuses on hard modality combinations rather than all possible combinations. Additionally, we introduce two baseline variations: (1) CMAD_w, which aligns with MMANet by considering difficult modality combinations but eliminates sample difficulty, and (2) CMAD_t, which considers all modality combinations without eliminating sample difficulty. To ensure a fair comparison, all KD-based models use the same teacher model, denoted as ‘TEA’.

Results on MOSEI. As shown in Table 1, the teacher model, trained using all modalities, exhibits significant feature inconsistencies and training instability, relying heavily on the language modality while paying insufficient attention to others. When language is available in a modality combination, the teacher model consistently outperforms most approaches. However, when the language modality is absent, its F1-score drops sharply, ranking below many competing models. Both MMANet and CMAD_w, designed to handle challenging modality combinations, significantly improve performance in difficult settings, such as {V} and {A, V}. However, they struggle to maintain the teacher model’s effectiveness in easier combinations. In contrast, our proposed CMAD framework not only preserves the teacher model’s performance in easy modality combinations but also substantially enhances results in more challenging settings, demonstrating robustness across all modality configurations. Furthermore, including sample difficulty from training leads to a decline in CMAD_t’s performance, falling below that of CMAD_w, underscoring the importance of incorporating sample difficulty. Although customized models achieve higher F1-scores than the teacher model, they fail to effectively balance the contribution of each modality, resulting in suboptimal overall performance.

Results on IEMOCAP. On the IEMOCAP dataset, the teacher model learns more consistent representations than on MOSEI. While language remains the dominant modality, its influence is less pronounced. MMANet and CMAD_w ex-

L	A	V	CMAD	w/o MAR	w/o CAFD
✓			86.1/86.0	85.3/85.3	85.6/85.4
	✓		63.0/ 60.8	58.9/59.5	63.8/60.4
		✓	65.7/64.4	63.8/63.0	63.3/62.6
✓	✓		86.3/86.2	85.1/85.1	85.5/85.4
✓		✓	86.4/86.4	85.6/85.5	85.4/85.3
	✓	✓	65.6/64.8	64.1/64.2	64.2/63.6
✓	✓	✓	86.1/86.1	85.6/85.6	85.7/85.6
Average			77.0/76.4	75.5/75.5	76.2/75.5

Table 2. Ablation results of modules on MOSEI dataset.

L	A	V	CMAD	w/o MAR	w/o CAFD
✓			79.5/78.7	78.6/77.2	78.9/77.8
	✓		79.2/76.7	77.9/73.0	79.2/75.9
		✓	73.9/ 70.6	75.3/70.1	72.8/69.2
✓	✓		81.9/81.3	80.6/79.5	81.4/80.4
✓		✓	79.7/79.1	79.7/78.6	79.0/77.9
	✓	✓	79.0/77.3	78.9/75.9	78.6/76.2
✓	✓	✓	81.7/81.2	81.5/80.6	81.5/80.4
Average			79.2/77.8	78.9/76.4	78.8/76.8

Table 3. Ablation results of modules on IEMOCAP dataset.

hibit similar trends as observed in MOSEI, improving performance in difficult modality configurations but underperforming in easier ones. When sample difficulty is ignored, performance declines as seen in MOSEI, with CMAD_t falling below CMAD_w. However, the full CMAD framework consistently enhances performance across nearly all modality configurations, maintaining robustness in diverse scenarios. Customized models show strong results in the {L, A, V} setting but fail to effectively utilize the unique contributions of each modality, leading to sub-optimal performance in other combinations.

Results on CHERMA. On the CHERMA dataset, CMAD consistently outperforms all other methods in all missing modality scenarios, achieving an average improvement of 1.4 over the second-best model, UMDF. Similar trends are observed in MOSEI and IEMOCAP, where the CMAD variant eliminating sample difficulty (CMAD_w) outperforms MMANet on average. In contrast, the CMAD variant including sample difficulty (CMAD_t) experiences a significant performance drop compared to the full CMAD framework. CMAD also achieves average performance improvements of 1.9% on MUSTARD and 0.5% on URFUNNY, which can be seen in Supplementary Material.

4.4. Ablation Studies

To better understand the contributions of the CAFD and MAR modules in CMAD framework, the ablation studies is conducted, as shown in Tables 2 and 3. Removing the CAFD module leads to a decline in performance across

nearly all model configurations on both the MOSEI and IEMOCAP datasets. Specifically, we observe an average drop in accuracy and F1-score of 0.8/0.9 on MOSEI and 0.4/1.0 on IEMOCAP. This decrease in performance can be attributed to the loss of information captured from both the representations and correlations through the teacher model, which is crucial for reducing the inconsistencies. Removing the MAR module, which involves eliminating the MAR weights from both $\mathcal{L}_{TASK}$ and $\mathcal{L}_{AUXI}$, leads to a greater drop in performance compared to the removal of the CAFD module. Specifically, the average accuracy and F1-score drop by 1.5/0.9 on the MOSEI dataset and 0.3/1.4 on the IEMOCAP dataset, respectively. Furthermore, even when only one component of the MAR weights (either from the $\mathcal{L}_{TASK}$ or $\mathcal{L}_{AUXI}$) is removed, there is still a noticeable decline in performance (details available in the Supplementary Material). These results demonstrate the effectiveness of the MAR weights in enhancing the model’s performance.

4.5. Qualitative Analysis

In this section, we provide additional analysis of the CAFD and MAR modules, with more details available in Supplementary Material. As shown in Table 4, we focus on analyzing the consistency within the CAFD module. Here $\mathcal{L}_{mse}$ and $\mathcal{L}_{sim}$ represent the feature consistency loss ($\mathcal{L}^{MSE}$), and correlation consistency losses ($\mathcal{L}^{S_p}$ and $\mathcal{L}^{S_a}$), respectively. A ‘✓’ indicates the usage of these consistency, while a ‘×’ denotes direct transfer of teacher-teacher correlations to the student-student correlations. Our findings reveal that: (1) Using only feature or correlation consistency independently reduces consistencies. (2) Transferring teacher-teacher correlations into student-teacher correlations—rather than directly applying teacher correlations to student representations—demonstrates better consistency across all modality combinations. (3) Integrating both feature and correlation consistency yields better and more consistent performance.

To further analyze the MAR weights, we visualized their progression on the MOSEI dataset before epoch 20, as shown in Fig. 3a, and compared the differences between the weights calculated with and without eliminating sample difficulty in Fig. 3b. In Fig. 3a, each value of MAR weights Ψ is computed using Eq. 10, based on accumulated values from previous epochs. As training progresses, the weights for easy modality combinations, particularly those involving language modality, decrease gradually, while weights for difficult combinations remain relatively high. This trend inversely correlates with modality performance, assigning higher loss weights to combinations with poorer performance. When sample difficulty is included, the general trend in MAR weights remains similar to Fig. 3a, but the range of weight adjustments becomes narrower. Weights for easy modality combinations decrease more slowly, sta-

Losses		Avg	
$\mathcal{L}_{mse}$	$\mathcal{L}_{sim}$	MOSEI	IEMOCAP
✓	✓	77.03/76.39	79.23/77.84
✓		76.43/75.71	78.36/76.61
	✓	76.39/75.64	78.84/77.27
	×	76.23/75.43	77.78/76.39

Table 4. Ablation results of consistency losses in $\mathcal{L}_{CAF D}$ on MOSEI and IEMOCAP datasets.

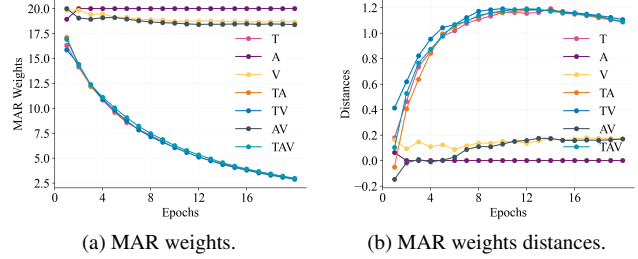


Figure 3. Visualizations of MAR weights and MAR weights distances on MOSEI dataset.

bilizing around 4, compared to around 2.5 when sample difficulty is eliminated (details available in the Supplementary Material). Furthermore, for easy modality combinations, the weight differences initially increase before decreasing, while weights for difficult combinations gradually increase. This leads to a shift in the relative magnitudes of MAR weights, from an 8-fold difference to a 4-fold difference. This pattern suggests that when sample difficulty is eliminated, the model learns a more consistent relationship between modality combinations, resulting in a more robust representation that adapts more effectively across various modality configurations.

5. Conclusion

In this paper, we present a unified framework, Correlation-Aware and Modalities-Aware Distillation (CMAD), to obtain unified and consistent representations for MSA with various missing modalities. CMAD incorporates a sample-wise consistency mechanism that guides the student model to learn sample-wise feature and correlation consistency from the teacher model. Additionally, the proposed adaptive weighting strategy helps the student model develop consistent representations by effectively adapting to diverse modality combinations, independent of sample difficulty. Extensive experiments on five datasets demonstrate that CMAD significantly outperforms existing methods. However, CMAD does not fully account for the varying contributions of each modality in the teacher model, which leads to a dominance effect by language modality. We will explore this in the future work.

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